

Smart Parking Assist - Pin Connection Sheet

ATmega2560 (Arduino Mega 2560) | Bare-metal drivers | Code: Pravinkumarvs/ATMEGA_2560 | Prepared by Ponmudi | 20 of 70 I/O pins used, no clashes

1. Power

From	To	Note
Mega 5V	Breadboard + rail	every VCC / 5V pin goes here
Mega GND	Breadboard - rail	every GND goes here (common ground)
-	-	Long breadboards: + and - rails may be split in the middle. Bridge the gap with a jumper.

2. Mandatory components

Component	Part pin	Connect to	Mega pin	Chip pin	Driver
HC-SR04 ultrasonic	VCC / Gnd	+ rail / - rail	-	-	ultra
	Trig	direct	D13	PB7	ultra
	Echo	direct	D12	PB6	ultra (Timer1)
IR sensor (slot)	VCC / GND	+ rail / - rail	-	-	ir
	AO (analog)	direct	A0	PF0	ir -> adc ch 0
Buzzer module	S	direct	D5	PE3 / OC3A	pwm (Timer3)
	middle / -	+ rail / - rail	-	-	-
Push switch	leg 1	+ rail	-	-	-
	leg 2	direct, and 10 kohm to - rail	D2	PE4	switch (pull-down)
LED	long leg	220 ohm	D11	PB5	led
	short leg	- rail	-	-	-
7-segment x1 (CC)	see section 4	1 kohm per segment line	D22-D29	PA0-PA7	segment

Fixed in the driver code: ultrasonic pins (*ultra.h*), buzzer pin D5 (*pwm.c*), LCD pins (*lcd.h*). **Chosen in the application:** IR channel *ir_init(ADC0)*, LED *led_init(PORT_B, 5)*, switch *switch_init(PORT_E, 4)*, 7-segment port *c_segment(PORT_A, digit)*. If the app uses other pins, change this sheet to match.

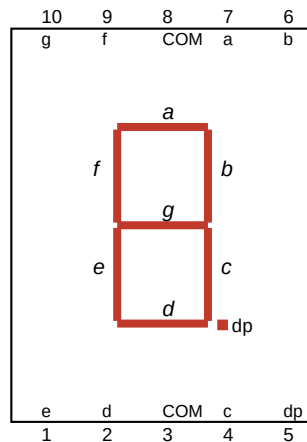
3. Timers

Timer	Job	Mode	Pins
Timer0	free	-	-
Timer1	delays (s / ms / us), ultrasonic echo timing (0.5 us per tick), LCD timing	CTC (mode 4) for delays, Normal (mode 0) for echo, polled	none
Timer2	free	-	-
Timer3	buzzer, 2 kHz	Fast PWM (mode 14), TOP = ICR3	D5 (OC3A)
Timer4	free (OC4A is D6, not D5)	-	-
Timer5	free	-	-

Timer1 is shared, so delays, ultrasonic measurement and LCD writes run one after another, never at the same time.

4. 7-segment display - one SUN056CC (common cathode)

The driver drives one display from **all 8 pins of one port**: pin 0 = a, 1 = b, ... 6 = g, 7 = dp. With *PORT_A* that is D22-D29. Use *c_segment()* for common cathode. For a common anode display (SUN056CA) use *a_segment()* and connect COM to the + rail instead.



Segments a-g, dp -> PA0-PA7
(D22-D29), 1 kohm each
COM (3 or 8) -> - rail (GND)

Hold the display with the decimal point at the bottom right. Pins 1-5 run along the bottom (left to right), pins 6-10 along the top (right to left).

Display pin	Segment	Resistor	Mega pin	Chip pin
7	a	1 kohm	D22	PA0
6	b	1 kohm	D23	PA1
4	c	1 kohm	D24	PA2
2	d	1 kohm	D25	PA3
1	e	1 kohm	D26	PA4
9	f	1 kohm	D27	PA5
10	g	1 kohm	D28	PA6
5	dp	1 kohm	D29	PA7
3 (or 8)	COM	none	- rail (GND)	-

If too dim, use 680 ohm. Do not go much lower: all 8 segments of digit 8 come from port A at the same time, and the ATmega2560 limits the total current of a port group to about 100 mA.

5. LCD 16x2 (JHD162A, 4-bit mode)

LCD pin	Connect to	Mega pin	Chip pin	Driver
1 VSS / 2 VCC	- rail / + rail	-	-	lcd
3 VEE (contrast)	10 kohm pot middle leg (outer legs to + and - rail), or about 1 kohm to - rail	-	-	-
4 RS	direct	A8	PK0	lcd
5 R/W	- rail (write only)	-	-	-
6 E	direct	A9	PK1	lcd
7-10 DB0-DB3	not connected (4-bit mode)	-	-	-
11-14 DB4-DB7	direct	A12, A13, A14, A15	PK4-PK7	lcd
15 LED+ / 16 LED-	+ rail via 220 ohm / - rail	-	-	-

Port K is also ADC8-ADC15, so the IR sensor must use A0-A7. **Blank screen** = contrast too low (turn the pot). **Top row of solid boxes** = LCD not initialised: check RS, E, DB4-DB7 and that R/W is on the - rail. **Garbage characters** = a data wire swapped or loose.

6. Pin usage check (sorted by port) - no pin used twice

Chip pin	Mega pin	Component	Signal
PA0	D22	7-segment	a
PA1	D23	7-segment	b
PA2	D24	7-segment	c
PA3	D25	7-segment	d
PA4	D26	7-segment	e
PA5	D27	7-segment	f
PA6	D28	7-segment	g
PA7	D29	7-segment	dp
PB5	D11	LED	long leg (via 220 ohm)
PB6	D12	Ultrasonic	ECHO
PB7	D13	Ultrasonic	TRIG
PE3 / OC3A	D5	Buzzer	S (PWM)
PE4	D2	Switch	leg 2 + 10 kohm to GND
PF0 / ADC0	A0	IR sensor	AO (analog)
PK0	A8	LCD	RS
PK1	A9	LCD	E
PK4	A12	LCD	DB4
PK5	A13	LCD	DB5
PK6	A14	LCD	DB6
PK7	A15	LCD	DB7

Total: 20 of 70 I/O pins used, 50 free. Checked automatically: no Mega pin and no chip pin appears twice.

7. Wiring checklist

- [] Power rails first, then test one part at a time.
- [] All modules share one GND with the Mega.
- [] LED has a 220 ohm resistor. Long leg towards the Mega pin.
- [] 7-segment is common cathode (SUN056CC): 1 kohm on each of the 8 segment lines, COM straight to the - rail.
- [] Switch: one leg to + rail, other leg to D2 and 10 kohm to - rail (the driver has no internal pull-up).
- [] IR module: use the analog output AO on A0. Adjust the blue pot until the module LED reacts at the right distance.
- [] Buzzer on D5 (not D6).
- [] LCD: connect VEE (pin 3) for contrast, otherwise the screen looks blank. R/W (pin 5) to the - rail.
- [] Upload: make flash PORT=/dev/ttyACM0